AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM341
Computer Vision

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 15/Nov./2025

Number of Exam Pages: 4 (including this cover sheet)	Time Allowed: 120 Minutes

Instructions:

- Total Marks: 30
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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Part – A (5 Marks)

1. The following code is for
   flipped_code_1=cv2.flip(input_image,1)
   a) Horizontal flip
   b) Vertical flip
   c) Transpose
   d) Translation

2. Closing is
   a) Dilation followed by Erosion
   b) Erosion followed by Dilation
   c) Dilation
   d) Erosion

3. Chane domain of image is an
   a) image warping
   b) image filtering
   c) image detection
   d) image display

4. In computer vision, in the high level stage the output is an
   a) image
   b) features
   c) analysis
   d) None of the above

5. Logarithmic transformation is useful to enhance the details in __ regions of an image.

   a) bright
   b) low-intensity
   c) high-frequency
   d) noisy

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Part – B (10 Marks)

Question 1: (3 marks)

a) What is the key difference between computer vision and image processing in terms of the intended use of their outputs?

b) Fill in the blanks to complete the Python code that applies a negative transformation to a grayscale image:

```
python
import cv2 as cv
import numpy as np
import matplotlib.pyplot as plt

Load the image in grayscale
img = cv.imread('image.jpg', cv.IMREAD_____)

Apply negative transformation
neg_img = _____ - img

Display the result
plt.imshow(neg_img, cmap='gray')
plt.title("Negative Image")
plt.axis('off')
plt.show()
```

Question 2: (4 marks)

a) Explain the purpose of applying a Gaussian filter in image processing.

b) You are given a 5×5 grayscale image (matrix of pixel values) and asked to apply a 3×3 Gaussian filter with the following kernel:

[[1, 2, 1],
[2, 4, 2],
[1, 2, 1]]

This kernel should be normalized (i.e., **divide all elements by the sum of the kernel values**).

c) Calculate the new value of the center pixel after applying the Gaussian filter. Assume the 3×3 region around the pixel is:

[[100, 100, 100],
[100, 150, 100],
[100, 100, 100]]

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Question 3: (3 marks)

Fill in the blank with the required code:

```
image = cv2.imread('image.jpg', cv.IMREAD_GRAYSCALE)

# Get image dimensions
a) _rows, cols=_____

# Define the translation matrix: move 200 pixels right and 100 pixels down
b)_____

# Apply affine transformation (translation)

c)_____
```

Part – C (15 Marks)

Question 1: (5 marks)

a) Explain the concept of filtering in the frequency domain. Why would we choose to filter an image in the frequency domain instead of the spatial domain?
b) What is the difference between a low-pass filter and a high-pass filter in the frequency domain? Give one application for each.
c) Write a python code to create a mask (low pass filter)

Question 2: (5 marks)

1. Compare Harris Corner Detection and SIFT in terms of feature detection. Provide a brief explanation.
2. Write a python code to
    1. detect and dilate corners using Harris
    2. apply threshold and mark corners red

Question 3: (5 marks)
a) Explain how the Sobel operator is used for edge detection.

b) Write a Python program that applies Sobel filters in both x and y directions on a grayscale image and displays the edge magnitude.

End of questions

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